Directed Low-Diameter Decomposition

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Let $G = (V, E, w)$ be a directed graph with nonnegative arc lengths. A *low-diameter decomposition* (LDD) removes a random set of arcs F such that every strongly connected component of $G - F$ has *weak diameter* at most D: for u, v in the same component, $d_G(u, v) \le D$, where the connecting path may leave the component. Its *loss* is the least ℓ satisfying

$$\Pr[e \in F] \le \min\left\{1, \frac{\ell w(e)}{D}\right\} \qquad (e \in E). \tag{1}$$

We obtain loss $O(\log n \log\log n)$ in expected time $O((m + n\log\log n)\log n)$ for polynomially bounded integral lengths.

Relation to prior work. This note simplifies the directed LDD construction of Li, Mowry, and Rao [1] and improves its expected construction time from $O((m + n\log\log n)\log n\log\log n)$ to $O((m + n\log\log n)\log n)$ for polynomially bounded integral lengths, while retaining the loss bound $O(\log n\log\log n)$.

An outward ball has short paths from its center, but need not have small directed diameter. We can nevertheless separate it from the rest of the graph by deleting only its outgoing boundary, then recurse inside it. There are two kinds of progress: a ball may be small enough to recurse on, or a region may have short paths in *both* directions and already satisfy the diameter bound. We use these alternatives to construct smaller instances, and charge the search work and cutting probabilities to the size reduction.

Discard self-loops and retain only the shortest arc of each ordered pair, assigning its cut decision to all parallel copies. Hence $m \le n^2$. If $D < 1$, delete every positive-length arc; below assume $D \ge 1$. We process initial SCCs independently and ignore isolated vertices.

Degree volume and one-sided carving. Fix a recursive instance H with M arcs. Throughout an attempt on H, use its original degrees to define

$$T = 2M, \qquad \mu(X) = \sum_{v \in X} \deg_H(v). \tag{2}$$

An arc internal to X contributes two to $\mu(X)$, and an arc crossing its boundary contributes one. Therefore

$$|E(H[X])| \le \frac{\mu(X)}{2}, \qquad |E(H[V(H) \setminus X])| \le M - \frac{\mu(X)}{2}. \tag{3}$$

Thus a small-volume set contains few arcs, while every component outside a large-volume set contains few arcs. Keeping degrees fixed as vertices are removed makes volume suitable for sampling as well.

Write $B_J^+(s, r) = \{v : d_J(s, v) \le r\}$ and $B_J^-(s, r) = \{v : d_J(v, s) \le r\}$. A *$\sigma$-pass*, $\sigma \in \{+, -\}$, fixes a graph snapshot J, some centers, and a common radius r. It processes the centers in random order. If U is

the current remainder, it carves $P = B_J^\sigma(s, r) \cap U$. For $\sigma = +$ it deletes arcs from P to $U \setminus P$; for $\sigma = -$ it deletes arcs from $U \setminus P$ to P. Then P is removed from U.

Only one boundary direction is deleted. In the outward case a surviving directed cycle cannot leave P; in the inward case it cannot enter P. Hence no SCC can meet both P and $U \setminus P$. Successive pieces and the final remainder can be treated independently. Throughout the pass distances are measured in the *fixed* graph J, even along paths through previously carved vertices. Moreover all sampled centers are processed, including those already carved. These conventions are essential: a center sampled inside a ball must still reach its target, regardless of earlier carving.

Lemma 1 (Random-order carving)**.** *Suppose a σ-pass uses k fixed centers in random order and a common radius uniform in $[a, b]$, where $a < b$. It cuts an arc e with probability at most $H_k w(e)/(b-a)$, where $H_k = \sum_{j=1}^{k} 1/j \le 1 + \ln(k+1)$. Its searches can be implemented so that the number R_v of scans at a vertex v satisfies*

$$\mathbb{E}R_v \le H_k, \qquad \mathbb{E}[2^{R_v}] \le k + 1. \tag{4}$$

A scan costs $O(\deg_J(v) + \log\log n)$ time.

Proof. Consider an outward pass and $e = u \to v$. A center s can cut e only if $d_J(s, u) \le r < d_J(s, v) \le d_J(s, u) + w(e)$, so the permissible radii have total length at most $w(e)$. Rank the centers by their distance to u, breaking ties consistently. If the center of rank j cuts e, none of the $j-1$ closer centers can have been processed first: at radius $r \ge d_J(s, u)$, any such center would already have carved u. The probability is therefore at most $1/j$, and summing over ranks proves the cutting bound. The inward case reverses the arcs.

For the searches, store at each vertex the smallest distance reached by an earlier center. If a new search reaches v at a distance no smaller than this record, it need not continue through v: every radius-r continuation would already have been reached by the earlier center. Thus a vertex is scanned only at record minima among the center distances. In random order, the record indicators are independent with means $1, 1/2, \ldots, 1/k$ (zero or infinite distances can only reduce the scan count). Hence, since for an indicator I we have $2^I = 1 + I$, $\mathbb{E}2^{R_v} \le \prod_{j=1}^{k}(1 + 1/j) = k + 1$. □

To sample centers at rate ρ, take $\lceil \rho \rceil$ independent draws from $\Pr[s = v] = \deg_H(v)/T$, discard duplicates and vertices outside the current snapshot, and use every nonisolated vertex instead if $\rho \ge T$. For any fixed set X in the snapshot,

$$\Pr[S \cap X = \varnothing] \le \left(1 - \frac{\mu(X)}{T}\right)^{\lceil \rho \rceil} \le e^{-\rho\mu(X)/T}. \tag{5}$$

One alias table, constructed in $O(M)$ time, supports all these draws in $O(1)$ time each.

The main use of the sampling bound is *opposite-direction coverage*. If $s \in B_J^+(u, r)$, then $u \in B_J^-(s, r)$, and conversely. Hence an inward pass of radius at least r removes u whenever it samples any vertex of $B_J^+(u, r)$; an outward pass similarly covers a large inward ball. In particular, a vertex whose ball in the opposite direction has volume greater than T/q survives with probability at most $e^{-\rho/q}$. The ball and the snapshot must be fixed before the centers of this pass are sampled.

Pivot, peel, and bootstrap. A decomposition attempt either uses a two-sided pivot immediately or peels small one-sided balls and then runs a sequence of CKR passes. We call the first pass after peeling the *bootstrap pass*; the subsequent, doubly exponential passes are numbered $1, 2, \ldots$. The bootstrap uses constant sampling and has a weaker size bound. Its purpose is to make the first numbered pass safe.

Draw K degree-weighted samples, which we call probes, for a sufficiently large constant K. For $\sigma \in \{+, -\}$, call s *σ-small* when $\mu(B_H^\sigma(s, D/4)) \le T/4$.

If some probe s is large in both directions, independently choose r_+, r_- uniformly from $[D/4, D/2]$ and cut the outgoing boundary of $B_H^+(s, r_+)$ and the incoming boundary of $B_H^-(s, r_-)$. Every resulting SCC is either contained in or disjoint from each ball. An SCC in their intersection has weak diameter at most D: the route $u \to s \to v$ has length at most $r_- + r_+ \le D$. Every other SCC avoids a ball of volume greater than $T/4$, so it contains fewer than $3M/4$ arcs by Equation (3). We accept this without further passes.

Otherwise every probe is small in at least one direction, so some σ has at least $K/2$ σ-small probes. Process their σ-balls in random order using one radius uniform in $[D/8, D/4]$. Each carved piece has volume at most $T/4$, although their union need not be small. On the remainder, run the *bootstrap pass*: for a

sufficiently large constant b we draw $\lceil b \rceil$ fresh degree-weighted centers, carve in direction $-\sigma$ at one radius uniform in $[D/12, D/8]$, and reject the entire attempt if any piece has volume exceeding $3T/4$.

The peel makes such a rejection unlikely. Fix u and a direction σ, and suppose $W = B_H^{-\sigma}(u, D/8)$ has $\mu(W) > 3T/4$. Any selected σ-small center in W has u in its radius-$D/8$ σ-ball and hence removes u. Thus, if u survives the peel in direction σ, at least $K/2$ of the original probes lie outside W, a set of sampling mass below $1/4$. Consequently,

$$\Pr[u \text{ survives the peel and } \mu(W) > 3T/4] \le \Pr[\mathrm{Bin}(K, 1/4) \ge K/2] \le e^{-K/12}. \tag{6}$$

A bootstrap piece can exceed $3T/4$ only if its center has such a large $(-\sigma)$-ball. Taking a union bound over both choices of σ and the $\lceil b \rceil$ fresh centers gives rejection probability at most $2\lceil b \rceil e^{-K/12}$.

Alternating multiscale passes. The bootstrap carves $(-\sigma)$-balls. Its purpose is to make large σ-balls unlikely among the surviving vertices, so the next pass can carve in direction σ. Repeating this argument lets us alternate directions, using just one CKR pass at each scale.

Throughout a pass, distances are still measured in the graph induced by the vertices present at its beginning, its snapshot. Centers are processed in random order, including centers assigned to earlier pieces during the same pass. Each piece is the intersection of its ball with the vertices still unassigned. We delete its outgoing boundary for an outward pass, or its incoming boundary for an inward pass. Consequently, no remaining SCC meets two different pieces.

For $M > 16$, set

$$\begin{aligned} &J = \lceil \log_2 \log_2 M \rceil - 1, \qquad s_i = 2^{2^i} \quad (1 \le i \le J), \\ &a_0 = D/12, \qquad \delta = \frac{D}{32(J+1)}, \qquad a_i = a_{i-1} - \delta. \end{aligned} \tag{7}$$

Pass i carves in the direction opposite to its predecessor, starting with direction σ after the bootstrap. It chooses a common radius uniformly in $[a_i, a_{i-1}]$ and requires every carved piece P to satisfy

$$\mu(P) \le T/s_i, \qquad T = 2M. \tag{8}$$

If this fails, we discard the entire attempt and resample. Since every continuing SCC inside P has at most $\mu(P)/2 \le M/s_i$ arcs, acceptance guarantees a logarithmic size reduction of at least $\log_2 s_i = 2^i$.

The radius intervals are decreasing: every radius in the preceding pass is at least every radius in the next pass. Their total width is less than $D/32$. This allows the preceding pass to eliminate potentially oversized centers of the next pass.

Lemma 2 (Coverage by the preceding pass)**.** *Suppose pass i carves in direction σ_i, and the preceding pass makes q independent draws in direction $-\sigma_i$. If the preceding pass has minimum radius at least a_{i-1}, then, for every fixed vertex u,*

$$\Pr\left[\begin{array}{c} u \text{ survives the preceding pass, and} \\ \mu(B_{H_i}^{\sigma_i}(u, a_{i-1})) > T/s_i \end{array} \right] \le e^{-q/s_i},$$

where H_i is the snapshot of pass i.

Proof. Fix the snapshot H_{i-1} before the preceding pass and consider

$$W = B_{H_{i-1}}^{\sigma_i}(u, a_{i-1}).$$

The next snapshot is an induced subgraph of H_{i-1}. Hence, if u survives and its ball in H_i has volume greater than T/s_i, then $\mu(W) > T/s_i$.

Every $s \in W$ satisfies $u \in B_{H_{i-1}}^{-\sigma_i}(s, a_{i-1})$. Sampling such an s in the preceding pass therefore removes u, unless it was removed already. This holds even if s was assigned to an earlier piece during that pass, since all sampled centers are processed using the fixed snapshot.

The set W is fixed before the preceding samples are drawn. Each draw belongs to W with probability $\mu(W)/T > 1/s_i$. Thus survival requires all q draws to miss W, and has probability at most

$$\left(1 - \frac{\mu(W)}{T}\right)^q \le e^{-q/s_i}.$$

□

We now choose the sampling rates. If pass i makes N_i fresh degree-weighted draws, the lemma and a union bound give

$$\Pr[\text{pass } i \text{ rejects}] \le N_i e^{-q_{i-1}/s_i}, \tag{9}$$

where q_{i-1} is the number of draws in the preceding pass. For a final pass that processes all remaining vertices, the same bound holds by a union bound over the at most T possible centers.

There are two competing requirements. With ρ_i sampled centers, random-order CKR scans a vertex $O(\log(\rho_i + 1))$ times in expectation. We want this to be $O(\log s_i)$, so it can be paid for by the size reduction. At the same time, the preceding sampling rate must make $\rho_i e^{-\rho_{i-1}/s_i}$ very small.

Both requirements are met by

$$\rho_i = A2^{i+1}s_i^2 = A2^{i+1}2^{2^{i+1}} \qquad (1 \le i \le J), \tag{10}$$

where A is a sufficiently large constant. Since $s_i = s_{i-1}^2$, for $i \ge 2$ we have

$$\frac{\rho_{i-1}}{s_i} = A2^i, \qquad \log(\rho_i + 1) = O(2^i), \qquad \Pr[\text{pass } i \text{ rejects}] = e^{-\Omega(2^i)}. \tag{11}$$

For pass 1, whose predecessor is the bootstrap with $\lceil b \rceil$ draws, the rejection probability is at most $\lceil \rho_1 \rceil e^{-b/4}$.

Normally pass i makes $\lceil \rho_i \rceil$ independent draws, ignoring centers absent from its snapshot. If $\rho_i \ge T$, it instead processes every nonisolated vertex of the snapshot in random order and terminates the sequence of passes. There are at most T such vertices, so this does not increase the work or the rejection bound. It also empties the remainder, since every remaining vertex is a center. Such a pass occurs by $i = J$: $2^J \ge \frac{1}{2}\log_2 M$ implies $s_J^2 \ge M$, and hence $\rho_J \ge T$.

Algorithm 1 Directed low-diameter decomposition

```
function DirectedLDD(H, D)
    if |E(H)| ≤ 16 then
        return the constant-size two-sided-pivot decomposition
    M ← |E(H)|; construct the sampler
    Set J, (s_i, ρ_i), (a_i) as above
    repeat
        Start with the original graph H and no deleted arcs
        Draw K probes and test their inward and outward radius-D/4 balls
        if some probe is large in both directions then
            Cut its inward and outward balls
            Mark SCCs inside their intersection terminal
        else
            Choose σ with at least K/2 σ-small probes
            Peel their σ-balls at a common radius in [D/8, D/4]
            Bootstrap: carve (−σ)-balls using ⌈b⌉ fresh draws and radius in [D/12, D/8]
            Reject if some bootstrap piece has volume > 3T/4
            for i = 1, ..., J while the attempt survives do
                Reverse the carving direction
                Choose the radius uniformly in [a_i, a_{i−1}]
                if ρ_i < T then
                    Carve using ⌈ρ_i⌉ fresh draws
                else
                    Carve using all remaining nonisolated centers in random order
                Reject if some carved piece P has μ(P) > T/s_i
                if ρ_i ≥ T then
                    break
    until the attempt is accepted
    Recurse on every nonterminal SCC of the accepted graph
    return the union of deleted arcs
```

Each attempt is discarded completely if a size test fails, including all tentative cuts. In the pivot case, an SCC inside the intersection has weak diameter at most D, using the route $u \to s \to v$ through the pivot.

Every other SCC omits one ball of volume greater than $T/4$ and therefore has fewer than $3M/4$ arcs. In the peel case, peeled pieces have volume at most $T/4$, bootstrap pieces at most $3T/4$, and numbered pieces satisfy Equation (8). Thus every nonterminal recursive child has at most $3M/4$ arcs.

For $M \le 16$, there are at most 32 nonisolated vertices. Choose a center in a nonterminal SCC and cut its independent inward and outward balls at radii uniform in $[D/4, D/2]$. Every SCC in their intersection is terminal, while each remaining nonterminal SCC omits the chosen center. At most 32 such pivots are needed, and the cutting probability of each arc is $O(w(e)/D)$.

Expected running time. The bootstrap rejects with probability at most $2\lceil b\rceil e^{-K/12}$, by the peel's sampling bound. Pass 1 rejects with probability at most $\lceil \rho_1 \rceil e^{-b/4}$, and later rejection probabilities are $e^{-\Omega(2^i)}$ by Equation (11). Choose A, then b, then K sufficiently large. Every attempt is accepted with probability at least $p = 3/4$.

Let η_i denote the rejection bounds for numbered passes, and put $\tau_i = \sum_{j \ge i} \eta_j$. Their rapid decay gives

$$\sum_{i \ge 1} 2^i \tau_i = O(1). \tag{12}$$

We will use this to pay for the work of attempts that are rejected after reaching an expensive pass.

Fix a vertex v and let $R_i(v)$ count its record-minimum Dijkstra scans in pass i, taking $R_i(v) = 0$ if the pass is not reached. There are at most $N_i = \min\{\lceil \rho_i \rceil, T\}$ centers, so the random-order record bound Lemma 1 gives $\mathbb{E}[2^{R_i(v)}] \le N_i + 1$. For a sufficiently large constant B, define the excess $Z_i(v) = (R_i(v) - B2^i)_+$. A tail sum yields

$$\mathbb{E}Z_i(v) \le 2(N_i + 1)2^{-B2^i} = e^{-\Omega(2^i)}, \qquad \sum_{i \ge 1} \mathbb{E}Z_i(v) = O(1). \tag{13}$$

Thus pass i uses at most $B2^i$ scans at v, apart from a summably small expected excess.

First consider an accepted attempt. If v enters a recursive child with M' arcs after pass j, its total scan allowance is $\sum_{i \le j} B2^i = O(2^j)$. But the size guard gives

$$\log_2 \frac{M}{M'} \ge \log_2 s_j = 2^j. \tag{14}$$

The pivot, peel, and bootstrap require only constant allowance and give a constant-factor size reduction. If v becomes terminal, its allowance is at most $O(2^J) = O(\log M)$. The excess has $O(1)$ expectation per accepted attempt, since conditioning on acceptance increases its expectation by at most $1/p$.

Next consider a rejected attempt. If pass i is reached, rejection can occur only in that or a later pass. Consequently,

$$\mathbb{E}\left[\mathbf{1}_{\text{reject}} \sum_{i \ge 1} R_i(v)\right] \le B \sum_{i \ge 1} 2^i \tau_i + \sum_{i \ge 1} \mathbb{E}Z_i(v) = O(1). \tag{15}$$

The initializer and bootstrap also contribute constant expected scans. Since each attempt succeeds with probability at least $3/4$, restarts increase these bounds by only a constant factor.

Accepted scan allowances telescope along each vertex's recursive chain through the decreases in $\log M$. Every continuing child has at most $3M/4$ arcs, so the $O(1)$ excess and initialization costs are paid at most $O(\log m)$ times. Hence every vertex has $O(\log m)$ expected scans in total.

There is also $O(M)$ non-Dijkstra work per attempt. The sampling rates grow geometrically, and sampling stops as soon as the rate reaches T; therefore only $O(M)$ centers are generated. Sampling takes $O(1)$ time per draw, and piece volumes are checked by summing parent degrees over disjoint assigned sets. Timestamped arrays avoid resetting vertex state between passes. Each scan of v costs $O(\deg_H(v) + \log\log n)$ using integer-priority-queue Dijkstra. Thus

$$\mathbb{E}[T_{\text{LDD}}] = O((m + n \log\log n) \log n). \tag{16}$$

Cutting loss. We need to account for the fact that an accepted attempt is conditioned on passing its size tests in our analysis. The same rejection tails control this conditioning.

Fix an arc e, and let $\mathcal{A}$ denote acceptance, with $p = \Pr(\mathcal{A}) \geq 3/4$. Let E_i be the event that pass i is reached with both endpoints of e still in its remainder. Conditional on the preceding passes, the random-order CKR bound is

$$\Pr[e \text{ cut in pass } i \mid E_i, \text{ preceding history}] \leq \alpha_i(e) := \frac{H_{N_i} w(e)}{\delta} = O\left(\frac{w(e)(J+1)2^i}{D}\right). \tag{17}$$

Here $H_k = \sum_{j=1}^{k} 1/j$; replacing the actual number of centers by its deterministic upper bound N_i is harmless.

Since an attempt reaching pass i can be rejected only in that or a later pass, $\Pr(E_i \cap \mathcal{A}^c) \leq \tau_i$. Therefore

$$\Pr[e \text{ cut in pass } i \mid \mathcal{A}] \leq \frac{\alpha_i(e)\Pr(E_i)}{p} \leq \alpha_i(e)\left(\Pr(E_i \mid \mathcal{A}) + \frac{\tau_i}{p}\right). \tag{18}$$

Summing the correction terms over all numbered passes gives

$$O\left(\frac{w(e)(J+1)}{D}\sum_{i\geq 1} 2^i \tau_i\right) = O\left(\frac{w(e)(J+1)}{D}\right).$$

For the main terms, let j be the last numbered pass to which e is exposed in an accepted attempt. Their total contribution is $O(w(e)(J+1)2^j/D)$. If e survives into a recursive child with M' arcs, Equation (14) charges this to

$$O\left(\frac{w(e)(J+1)}{D}\log\frac{M}{M'}\right).$$

Otherwise recursion stops charging e, and $2^j \leq 2^J = O(\log M)$. The pivot, peel, and bootstrap contribute $O(w(e)/D)$ per invocation, including conditioning on acceptance.

The logarithmic size decreases telescope along the recursive chain of e. Every continuing child contains at most $3M/4$ arcs, so the conditioning correction is charged at most $O(\log m)$ times. Since $J = O(\log\log m)$, we obtain

$$\Pr[e \in F] = O\left(\min\left\{1, \frac{w(e)\log n \log\log n}{D}\right\}\right). \tag{19}$$

An arc of length zero is never cut: every outward ball is closed under outgoing zero-length arcs, and every inward ball is closed under incoming zero-length arcs.

Theorem 3 (Directed low-diameter decomposition)**.** *For nonnegative polynomially bounded integral arc lengths, a directed low-diameter decomposition of weak diameter at most D and loss $O(\log n \log\log n)$ can be constructed in expected time*

$$O((m + n\log\log n)\log n).$$

References

[1] Jason Li, Connor Mowry, and Satish Rao. *Faster Negative-Weight Shortest Paths and Directed Low-Diameter Decompositions*. 2025. arXiv:2510.22721v1.